

COEP Technological University, Pune
Department of applied sciences and humanities(Physics)

Qunatum Physics syllabus
Even semester 2025-26

Slot 1 syllabus

Unit	Contents	lectures
1	Mathematical tools for Quantum Physics: Review of linear vector spaces, Hilbert space, Dirac Notations(ket and bra vectors). Operators in Hilbert space: definition, Hermitian Ad-joint, projection operators, functions of operator, unitary operators, eigenvalues and eigenvectors of an operator. Commutator and anti-commutator bracket algebra. Representation in Discrete Bases: column and row matrix representation of ket and bra vectors, $N \times N$ square matrix representation of operators, change of bases and unitary transformations.	7
2	Postulates of quantum mechanics and simple harmonic oscillator problem: Basic postulates of quantum mechanics, the state of a system: probability density, superposition principle. Relation between observables and operators, measurement in quantum mechanics, expectation values, time-independent Schrodinger equation. Harmonic oscillator: energy eigenvalue and eigenstates of the Hamiltonian operator, raising and lowering operators(ladder operators), occupation number operator. Matrix representation of various operators.	7

Suggested resources for reading and exercises:

1. Quantum Mechanics: Concepts and Applications, 2nd edition by N. Zettili, Wiley Pub
2. Introduction to Quantum Mechanics, 2nd Edition, by David J. Griffiths